function of metric and its first two derivatives; (iii) it is divergence-less. Having determined the most general form of E_{ab} satisfying these criteria, he obtained a Lagrangian from which one can obtain these field equations. (Note that we have not demanded E_{ab} to be *linear* in the second derivatives of the metric. If such a restriction is *also* added, one obtains Einstein's theory with a cosmological constant.) It is, however, possible to obtain Lanczos-Lovelock models in a more streamlined and transparent manner along different lines (see e.g., Chapter 15 of Ref. [6]) which is what we will pursue here.

2.1 Lagrangians from curvature and metric: general results

Let us consider a *general* Lagrangian built out of Riemann tensor and the metric, but not containing derivatives of the Riemann tensor, and ask the question: What is the condition on such a Lagrangian so as to ensure that equations of motion should contain no derivatives of the metric tensor higher than second order? The answer, as it turns out, uniquely picks out the Lanczos-Lovelock Lagrangians.

We will now describe this result highlighting the main conceptual points, many of which are of interest in their own right. We start with the action functional in $D-$dimensional spacetime:

$$A = \int_{\mathcal{V}} d^D x \sqrt{-g}\, L \tag{1}$$

where L is a scalar built from the metric and the curvature tensor. We can think of the Lagrangian L as a functional of any one of $(R_{abcd}, R^a{}_{bcd}, R^{ab}_{cd}...)$ in combination with either g^{ab} or g_{ab}. The variation of the Lagrangian with respect to curvature and metric will bring in tensors defined by the derivative of Lagrangian with respect to different kinds of curvature components and the metric, like, for e.g.,

$$P^{abcd} \equiv \left(\frac{\partial L}{\partial R_{abcd}}\right)_{g_{ij}}; \qquad P^{ab} \equiv \left(\frac{\partial L}{\partial g_{ab}}\right)_{R_{ijkl}} \tag{2}$$

It is obvious from the above definition that P^{ab} is symmetric and P^{abcd} has the algebraic properties of the curvature tensor:

$$P^{abcd} = -P^{bacd} = -P^{abdc}; \quad P^{abcd} = P^{cdab}; \quad P^{a[bcd]} = 0 \tag{3}$$

Curiously enough, the *mere fact that L is a scalar imposes several further relations* among these derivatives which play a crucial role in deriving the Lanczos-Lovelock field equations [7]. The most important ones among these relations can be expressed in terms of the tensor

$$\mathcal{R}^{ab} \equiv P^{aijk} R^b{}_{ijk} \tag{4}$$

which plays the role analogous to Ricci tensor of Einstein's theory. It can be shown (see [7]; Appendix A summarises these results with proof) that the following relations hold: To begin with, P^{ab} and $\mathcal{R}^{ab}$ are closely related to each other:

$$\left(\frac{\partial L}{\partial g_{ab}}\right)_{R_{ijkl}} = -2R^b{}_{ijk}\left(\frac{\partial L}{\partial R_{aijk}}\right)_{g_{ab}}; \quad \text{i.e., } P^{ab} = -2\mathcal{R}^{ab} \tag{5}$$

It immediately follows that $\mathcal{R}^{ab}$ is symmetric; this is by no means an obvious result and does *not* follow from the algebraic properties of curvature tensor. Using Eq. (5), one can obtain several other relations connecting the partial derivatives when (g^{ab}, R_{abcd}) or $(g^{ab}, R^a{}_{bcd})$ or (g^{ab}, R^{ab}_{cd}) are used as independent variables to describe the system. If we use the pair (g^{ab}, R_{abcd}) as independent variables, we have

$$\left(\frac{\partial L}{\partial g^{im}}\right)_{R_{abcd}} = -\left(\frac{\partial L}{\partial g_{im}}\right)_{R_{abcd}} = 2\mathcal{R}_{im} \tag{6}$$

On the other hand, if we use the pair $(g^{ab}, R^a{}_{bcd})$ as independent variables, one can show that

$$\left(\frac{\partial L}{\partial g^{im}}\right)_{R^a{}_{bcd}} = \mathcal{R}_{im} \tag{7}$$

and, more interestingly, if we use the pair (g^{ab}, R^{ab}_{cd}) as independent variables, we get

$$\left(\frac{\partial L}{\partial g^{im}}\right)_{R^{ab}_{cd}} = 0 \tag{8}$$

Finally, it can be proved that the tensor $\nabla_m \nabla_n P^{amnb}$ is symmetric in a, b.

An *intuitive* way of understanding these results is as follows: To construct scalar *polynomials* of arbitrary degree in the curvature tensor, using only the curvature tensor R_{abcd} and metric g^{ij} (with index placements as chosen), we need two factors of the metric tensors to contract out the four indices of each curvature tensor in each term. So Eq. (6) is clearly true if L is a polynomial in R_{abcd}. Any arbitrary scalar function of curvature and metric can be represented as a (possibly infinite) series expansion in powers of the curvature tensor. Because Eq. (6) is linear in L, if it holds for a polynomial of arbitrary degree in curvature tensor, then it should hold for arbitrary scalar functions which possess a series expansion. This (intuitive) argument also gives the result in Eq. (8) which states that in scalar polynomials constructed from R^{ab}_{cd} the metric cannot appear and all the contractions must be with Kronecker deltas! This is obvious because, if we use g^{ab} to contract any two lower indices, that will leave two upper indices hanging loose which cannot be contracted out because we do not have g_{ab} available as an independent variable. Once we have Eq. (8) the other results can be obtained from it quite easily. Of course, the argument based on polynomials does not capture the full generality of our results, which hold for arbitrary Lagrangians that are not necessarily polynomials.

Having obtained these useful identities, let us now proceed with the variation of the action treating the Lagrangian as a function of R_{abcd} and g^{ab} with the index placements as shown. This leads to the result:

$$\delta A = \delta \int_{\mathcal{V}} d^D x \sqrt{-g}\, L = \int_{\mathcal{V}} d^D x\, \sqrt{-g}\, E_{ab} \delta g^{ab} + \int_{\mathcal{V}} d^D x\, \sqrt{-g}\, \nabla_j \delta v^j \tag{9}$$

where

$$\begin{aligned} E_{ab} &\equiv \frac{1}{\sqrt{-g}} \left(\frac{\partial \sqrt{-g} L}{\partial g^{ab}}\right)_{R_{abcd}} - \mathcal{R}_{ab} - 2\nabla^m \nabla^n P_{amnb} \\ &= \mathcal{R}_{ab} - \frac{1}{2} g_{ab} L - 2\nabla^m \nabla^n P_{amnb} \end{aligned} \tag{10}$$

and

$$\delta v^j \equiv [2P^{ibjd}(\nabla_b \delta g_{di}) - 2\delta g_{di}(\nabla_c P^{ijcd})] \tag{11}$$

In arriving at the second equality in Eq. (10) we have used Eq. (6). Note that each of the three terms in E_{ab} in Eq. (10) is individually symmetric in a, b.

In order to obtain sensible field equations from $\delta A = 0$, we need to ignore the boundary term involving δv^i. From Eq. (11) it is clear that both the variation of the metric and its first derivative normal to the boundary contributes at the boundary. In other words, we do *not* get a sensible variational principle if we only assume $\delta g_{ab} = 0$ at the boundary. (This problem, of course, is well known even in Einstein's theory.) To obtain a well defined action principle, we have to add some boundary terms to our action such that the variations of the normal derivatives of the metric are cancelled out. We will say more about it in Sec. 2.6 but will assume, for the moment, that the boundary term involving δv^i can be ignored or cancelled. In that case, taking the total action to be $A_{tot} = A + A_{matter}$ and defining the energy momentum tensor for matter from the variation of A_{matter} with respect to the metric in the usual manner, our field equations will reduce to $E_{ab} = (1/2)T_{ab}$ where T_{ab} is the matter energy momentum tensor.

2.2 Lanczos-Lovelock action functional

Since P^{abcd} involves second derivatives of the metric, the term $\nabla^m \nabla^n P_{amnb}$ in Eq. (10) will contain up to fourth derivatives of the metric. Hence, to satisfy our requirement that equations of motion should not have derivatives higher that second, we need to impose the condition:

$$\nabla_a P^{abcd} = 0 \tag{12}$$

(It turns out that the seemingly more general condition, $\nabla^m \nabla^n P_{amnb} = 0$ does not lead to anything different.) Due to symmetries of P_{abcd}, this makes P^{abcd} divergence-free in all indices. In this case, the equations of motion become

$$\mathcal{R}_{ab} - \frac{1}{2} g_{ab} L = \left(\frac{\partial \sqrt{-g} L}{\partial g^{ab}}\right)_{R^a{}_{bcd}} = \frac{1}{2} T_{ab} \tag{13}$$

In arriving at the first equality we have used Eq. (7). Clearly, the field equations do not involve more than second derivatives of the metric but are, in general, nonlinear functions of the second derivatives (the only exception being the Einstein-Hilbert Lagrangian). Note that the structure of E_{ab} is very similar to that in Einstein's theory with $\mathcal{R}_{ab}$ playing the role analogous to Ricci tensor.

Our original task of finding an action functional, leading to equations of motion which contain no derivatives higher than the second order, now reduces to finding all scalar functions of curvature and metric such that Eq. (12) is satisfied. We shall now describe how one such class of Lagrangians can be constructed but will not bother to prove that our construction is actually unique — for which we refer the reader to original literature [1, 2].

For this purpose, let us consider a general class of Lagrangians of polynomial form in curvature:

$$L = \sum_{m=0}^{M} L_m \left[(\mathrm{Riem})^m\right] \tag{14}$$